

System Driven Hardware Design

3-Bit Counter

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Agenda

- Task
- How to Proceed
- 3Bit-Counter
- Debouncing
- Helpful Resources

You have to set-up a 3-Bit Counter. The state of the counter is represented by the three LEDs: green/yellow/red. The counter is increased by pressing the push button and wraps around, when the maximum counts are reached.

Video: [How it should work at the end.](#)

You are experienced with the PSoC Creator:

Start the 3-Bit counter project directly, if you can answer the following questions:

- I know how to handle interrupts
- I know about the interrupt service routine
- I know how to use global flags to handover the interrupt to the main program
- I know bitwise XOR, AND and Bitwise shifts



Directly start with the 3-Bit Counter.

You are **not** experienced with the PSoC Creator:

Before you start with the 3-Bit Counter, do the following exercises:

1. Turn the user LED on the FreeSoc Board on by pressing the user Button
 - Read the FreeSoc Schematic for LED and Button Pinning
 - Get familiar with the interrupts in PSoC (see ISR Component Datasheet)
 - Look at the example Codes: Digital Pins
2. Turn the user LED on/off by pressing the user button
 - Learn how to toggle the LED status by pressing the button.
3. Start with the 3-Bit Counter

3-Bit Counter

Step-by Step

- Enter the Top-Schematic Components to be used (Pins, Push button, Leds, resistors, ISR Block)
- Configure your Pins and the ISR Block on the Top Schematic
- Calculate the Resistors in Front of you LEDs
- Check the VDDIO Switch positions of your FreeSoc2. They should be at 5 Volts
- Bread Board the LEDs, Resistors, Push Button and Capacitor
- Do the pin mapping
- Write the program Code (good to break it down in little task to test and debug it)
- Enter the ISR Code (Don't forget to clear the interrupt: `Pin_ClearInterrupt()`)
- Program the Device
- Start Debugging

Software Constraints

- The ISR should set a global flag, to show the Button was pressed (as the minimum, if you are familiar with other techniques, use them)
- Use a counter variable in `main()` to count the push button pressings
- Use a function to set the status LEDs (green, red, yellow) on and off

Hardware Constraints

- Use the three external LED with current limiting resistor (green, yellow, red)
- Use the external push button with 100nF capacitor in parallel to debounce it

Your task:

Implement a HW or SW Debouncing for your Push Button

Cypress Documentation (to be found in moodle)

PSoC® Creator™ Component Datasheet: **Interrupt 1.70**

PSoC® Creator™ Example Project: **Digital Pins 1.10**